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Shock dynamics and neutron production in an explosive generator driven dense plasma focus

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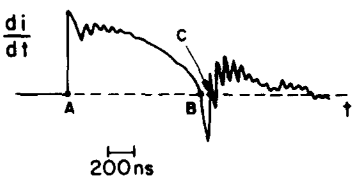

FIG. 4. Typical *di/dt* curve. Energy is transferred to the pinched plasma between B and C.

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$$\eta = \frac{\frac{1}{2}L_B I_B^2 - \frac{1}{2}L_c I_c^2}{\frac{1}{2}L_B I_B^2} = \frac{I_B - I_c}{I_B}.$$

Here, I_B is the current just before the pinch and I_c is the current just after the pinch. The efficiency can thus be deduced from the *di/dt* curve (Fig. 4). The efficiency for the 90% D_2-10% Ar mixture was $\eta_{D_2/Ar} \sim 6\%$ while for pure deuterium $\eta_{D_2} \sim 3\%$, a factor of 2 lower.

In conclusion, (1) the addition of a small percentage of argon results in a more uniform, efficient, and stable pinch. (2) The 90% D_2-10% Ar mixture results in $(N\tau)_D \sim 2\times 10^{12}$, an order of magnitude higher than for pure D_2. The optimum mixture has not yet been determined and further increase in $N\tau$ may be possible. (3) The neutron yield obtained with the present device compares favorably with the results obtained in a plasma focus. The yield from a plasma focus scales according to $Y \propto E^{2.5}$.[6,7] Typical yield for a plasma focus operating at 93 kJ is $Y \sim 3\times 10^{10}$ N/pulse.[6] Scaling this to an energy comparable to our device, we obtain $Y \sim 2\times 10^{7}$ N/pulse. However, in the low density regime the obtained neutron yield is an order of magnitude higher. (4) Further studies of the separation effect and a self-consistent model which will explain it, as well as several other mixtures and the resulting effects on column stability, temperature, $N\tau$, and neutron yield are under way.

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Shock dynamics and neutron production in an explosive generator driven dense plasma focus

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An explosive generator driven dense plasma focus is simulated numerically using a magnetohydrodynamic model which includes thermal conduction, resistive diffusion, and radiation in addition to the Lorentz force and shock hydrodynamics. It is shown that the dominant heating mechanism is shock heating. Neutron yield is shown to occur at the current minimum. The neutron yield is a result of the initial shock reflecting off the axis of symmetry, reflecting off the magnetic piston driving the focus, and reconverging on axis. Peak neutron production occurs when post-shock plasma converges on axis.

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The large scale numerical modeling of the dense plasma focus has been of interest since the pioneering work of Potter[1] and D'yachenko and Imshennik.[2] Potter's calculations demonstrated three features of the focus discharge: an anode cold source, a hot pinch region, and an axial shock. Most recently, Maxon and Eddleman[3] described the behavior of a focus with a rounded, hollow anode. Although D'yachenko and Imshennik and Maxon and Eddleman question the quantitative accuracy of Potter's results, the qualitative description provided by Potter remains valid. Neither Potter nor Maxon and Eddleman actually compute the neutron production rate and correlate it with the current waveform and the shock dynamics of the focus, although Potter gives an estimate.

We have recently initiated computations of the Los Alamos explosive generator driven dense plasma focus experiment.[4] In these experiments, an explosive driven magnetic flux compressor[5] replaces the traditional capacitor bank, leading to currents of several megamperes. Characteristic parameters of these experiments are anode diameter, 10.16 cm; cathode diameter, 15.24 cm; anode length, 20.0 cm; fill pressure, 22 Torr deuterium.

In our computations of the plasma behavior, we use a "two-temperature" magnetohydrodynamic model which includes electron and ion thermal conduction, resistive diffusion, and Bremsstrahlung radiation in addition to the Lorentz force and shock hydrodynamics. The magnetohydrodynamic model is solved self-consistently with an external circuit equation which represents the explosive generator by a preprogrammed inductance versus time profile which has been determined from generator experiments. The neutron yield is computed using Maxwell averaged cross sections for the *D-D* reaction.

The neutron yield from various plasma focus experiments has been attributed to both thermal mechanisms and to a nonthermal, beam-target interaction. There are basically two modes of plasma focus operation: low pressure and high pressure.[6,7] In the high pressure mode the thermal neutron-producing mechanisms are emphasized and become progressively more dominant as the power level of the experiment is increased.[8] The measurements of Trusillo *et al.*[9] suggest the dominance of the thermal mechanism. Because the Los Alamos plasma focus is operated in the high pressure mode, we would expect that magnetohydrodynamic computations such as reported here will give useful insight into the neutron-producing dynamics of the experiment.

Our computations do not include the polarity dependent effects shown by Potter to be essentially negligible. In addition, our model does not include ion viscosity. Rather, our calculations include an "artificial viscosity[10]" which provides for the correct increase in entropy when a shock traverses the plasma by irreversibly converting kinetic energy to ion thermal energy. Because the finite-difference mesh used in the computations limits the actual velocity gradients which can occur, it is conjectured that the "artificial viscosity" heating, or "shock" heating, would exceed the ion viscous heating should the latter be included. We note that the artificial viscosity is only artificial in the sense that it changes equations which are valid on either side of a shock, but not through a shock, into equations which are valid through a shock when the finite-difference mesh will not allow the steep gradients required for real viscosity effects to be important. Failure to include an artificial viscosity in finite-difference calculations can lead to either incorrect entropy production or disappearance of energy.[11]

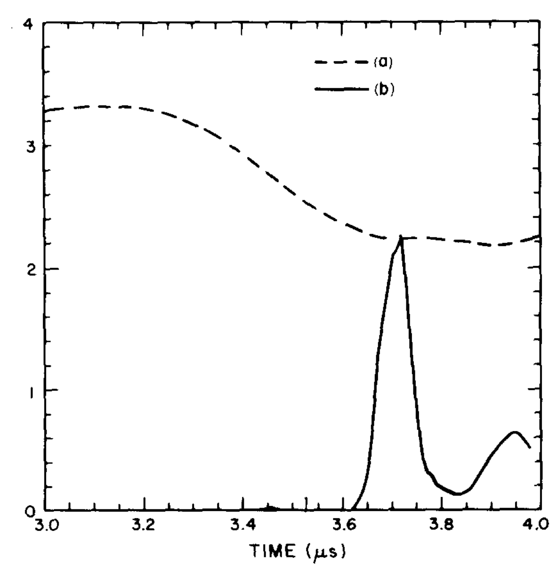

FIG. 1. (a) Current, 10^6 A; (b) neutron production rate, 10^{10} neutrons/μs.

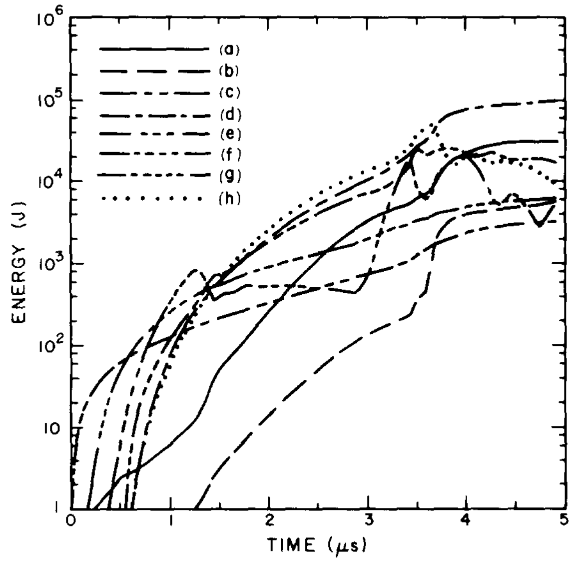

FIG. 2. Plasma energetics: (a) electron thermal conduction loss; (b) ion thermal conduction loss; (c) radiation loss; (d) shock heating; (e) ion or electron $p\nabla\cdot\mathbf{v}$ heating; (f) Ohmic heating; (g) radial kinetic energy; (h) axial kinetic energy.

In our calculations completed to date, we have observed features of the dense plasma focus discharge which have not been examined in detail by previous researchers. We have found that shock heating is the dominant heating mechanism, and, consequently, the electrons are heated primarily by ion-electron coupling rather than by Ohmic heating. At the densities of the generator driven experiments, the ions and electrons remain at essentially the same temperature except during the time shock waves are converging on axis. Most notably, we have found that the peak neutron production begins not at the initial focus when a first shock reaches the axis but rather when the initial reflected shock is itself reflected off the magnetic piston and a repinch occurs.

The computed current signal and neutron production signal is shown in Fig. 1. The peak neutron rate occurs at 3.7 μs, corresponding to the leveling of the current waveform when the discharge inductance reaches a maximum. Another neutron rate peak occurs at approximately 3.95 μs and is the result of axially moving shocks being reflected off the anode and off an endplug artificially located 10 cm beyond the anode for computational convenience. Barely perceptible in Fig. 1 is an initial rate peak at 3.45 μs which results when the initial radial shock converges on axis.

The plasma cooling and heating mechanisms and the kinetic energies are displayed in Fig. 2. Because our computations assume an initial temperature of 1 eV, the plasma initially cools by radiation. However, for most of the discharge, electron thermal conduction is the dominant loss mechanism. The computation displayed in Fig. 2 includes only Bremsstralung, but because during the rundown stage the plasma temperature is 15–30 eV, it would be expected that bound-free and bound-bound radiation is important. When we included bound-free and bound-bound radiation in our calculations, we found that the radiation losses increased by a factor of approximately 5, suggesting that radiation could become dominant at higher fill pressures.

Figure 2 shows that shock heating exceeds Ohmic heating by over an order of magnitude once the discharge is underway. Because the ions and electrons are essentially in thermal equilibrium, the $p\nabla\cdot\mathbf{v}$ heating is the same for both. The shock heating which occurs to the ions is distributed to the electrons by ion-electron coupling.

After the initial liftoff from the insulator which is located on the anode, the axial kinetic energy exceeds the radial kinetic energy. When the discharge reaches the end of the anode at about 3 μs, the radial kinetic energy abruptly increases until the radial imploding shock reaches the axis. At that time, the shock and $p\nabla\cdot\mathbf{v}$ heating increase abruptly, accompanied by an increase in axial kinetic energy resulting from the generation of axial shocks.

Figure 3 gives the electron density history at four points along the axis beyond the anode. At each point, an initial compression is followed by an expansion and subsequent recompression. The plasma history at the actual focus located 1.75 cm from the anode is shown in Fig. 4. The peak temperature which occurs in the focus region during the initial compression is 370 eV, but during the recompression the electrons reach 2 keV and the ions reach 3.7 keV. The peak neutron production rate actually occurs when the density on axis has decreased to a value within a factor of 3 of the initial value. The plasma in the focus at the time of neutron production is not the plasma originally located on axis, but rather post-shock plasma converging on axis, hence the temperature increases even though the density decreases. Axial flow both away from and toward the anode is essential to remove the plasma originally located in the focus region. We have observed plasma being radially ejected along the anode as a result of an axial shock moving from the focus toward the anode. Plasma behavior qualitatively similar to that depicted in Figs. 3 and 4 has been computed by Vikhrev and Korzhavin[12] using a one-dimensional model which included end effects and by D'yachenko and Imshennik.[2] The complex details of the flow pattern leading to the peak temperature well after peak density occurs are a subject of continuing analysis.

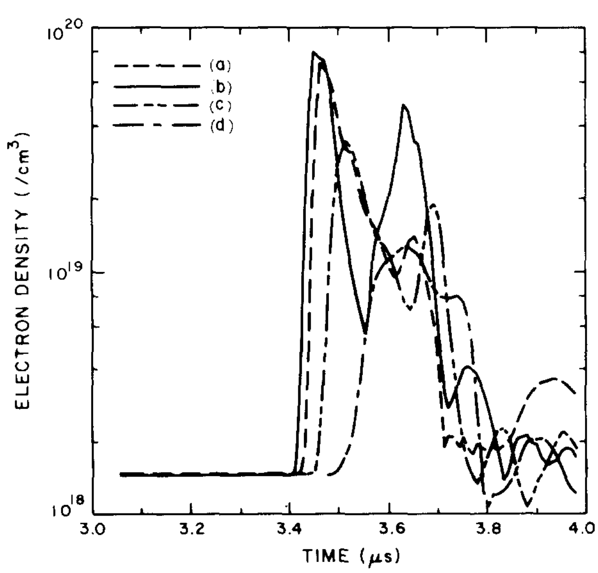

FIG. 3. Electron density for points located on axis: (a) at the anode; (b) 2.5 cm from the anode; (c) 5.0 cm from the anode; (d) 7.5 cm from the anode.

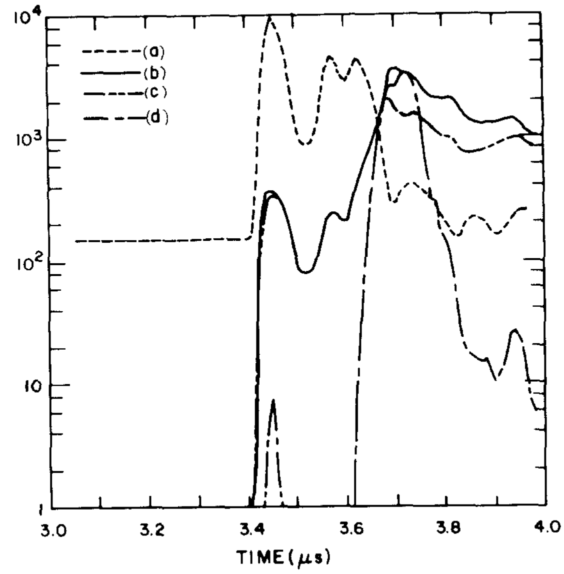

FIG. 4. Focus history: (a) electron density, 10^{16}/cm^3; (b) ion temperature, eV; (c) electron temperature, eV; (d) *D-D* reaction rate, 10^8 neutrons/(cm^3 μs).

Our computations are in qualitative agreement with preliminary data from actual experiments which have been performed at this laboratory.[8] The time at which the current peaks, the time at which the peak neutron yield occurs with respect to the current, and the neutron pulse full width at half-maximum agree with the experimental observations. The actual value of the peak current in the computations is somewhat high, suggesting a problem with circuit parameters. The location of the focus is too far from the anode, the number of neutrons produced is too low, and the peak temperatures achieved are too low; these differences are attributed to inadequate computational zoning in the vicinity of the focus. The spatial resolution of our computations completed to date is not sufficient to preclude possible contributions of a nonthermal mechanism to the experimental neutron yield.

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